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Assignment Title	Comprehensive Data Exploration, Analysis and Visualization Using Pandas and Matplotlib
Course Outcome(s) – CO	CO1: Apply data-importing, exploration, cleaning, grouping, joining and analytical techniques using Python and Pandas to extract meaningful information from datasets. CO2: Analyse and visualize data using Pandas and Matplotlib to identify relationships, correlations, distributions, outliers and time-based patterns for effective data-driven decision-making.
Bloom's Taxonomy Level	BL5 – Evaluate
SDG Mapping (SDG 1–17)	SDG 4 – Quality Education: Develop practical knowledge, analytical skills and problem-solving abilities in data exploration and visualization. SDG 9 – Industry, Innovation and Infrastructure: Apply data analytics and visualization techniques to support data-driven decision-making and technological innovation.

Comprehensive Data Exploration, Analysis and Visualization Using Pandas and Matplotlib

1. Introduction

Data analytics involves collecting, processing, analysing and visualizing data to extract useful information for decision-making. In this assignment, Pandas and Matplotlib are used to perform data importing, exploration, cleaning, joining, grouping, filtering, statistical analysis and visualization.

The project works with sales and customer datasets. Different analytical techniques are applied to

identify trends, relationships, distributions, correlations and outliers. Various visualization techniques are also implemented to understand the data more effectively.

2. Problem Statement

A data analytics organization receives large datasets from multiple sources and needs to transform raw data into meaningful information. The objective of this assignment is to demonstrate a systematic process for importing, exploring, cleaning, combining, analysing and visualizing datasets using Python, Pandas and Matplotlib.

The analysis focuses on identifying important patterns, relationships, trends, distributions and outliers while presenting the results using suitable visualizations.

This directly follows the problem statement given in your assignment.

3. Objectives

The main objectives of this assignment are:

- To import datasets using Pandas.
- To explore the structure and characteristics of datasets.
- To handle missing values and clean the data.
- To join multiple datasets.
- To perform grouping and filtering operations.
- To calculate descriptive statistics.
- To identify correlations between numerical variables.
- To detect outliers using statistical techniques.
- To create different visualizations using Matplotlib.
- To implement advanced visualization techniques.
- To analyse time-based trends.
- To evaluate the suitability of different visualization techniques.

4. Tools and Technologies

Tool	Purpose
Python	Programming language
Pandas	Data manipulation and analysis
NumPy	Numerical operations and dataset generation
Matplotlib	Data visualization
CSV	Dataset storage format
VS Code / IDLE / PowerShell	Program execution

5. Dataset Description

The project uses two related datasets:

Sales Dataset

The sales dataset contains information such as:

- Order ID
- Customer ID
- Date
- Product
- Category
- Quantity
- Unit Price
- Discount
- Rating
- Sales Amount

Customer Dataset

The customer dataset contains:

- Customer ID

- Customer Name
- City

The datasets are generated using Python for educational and reproducibility purposes.

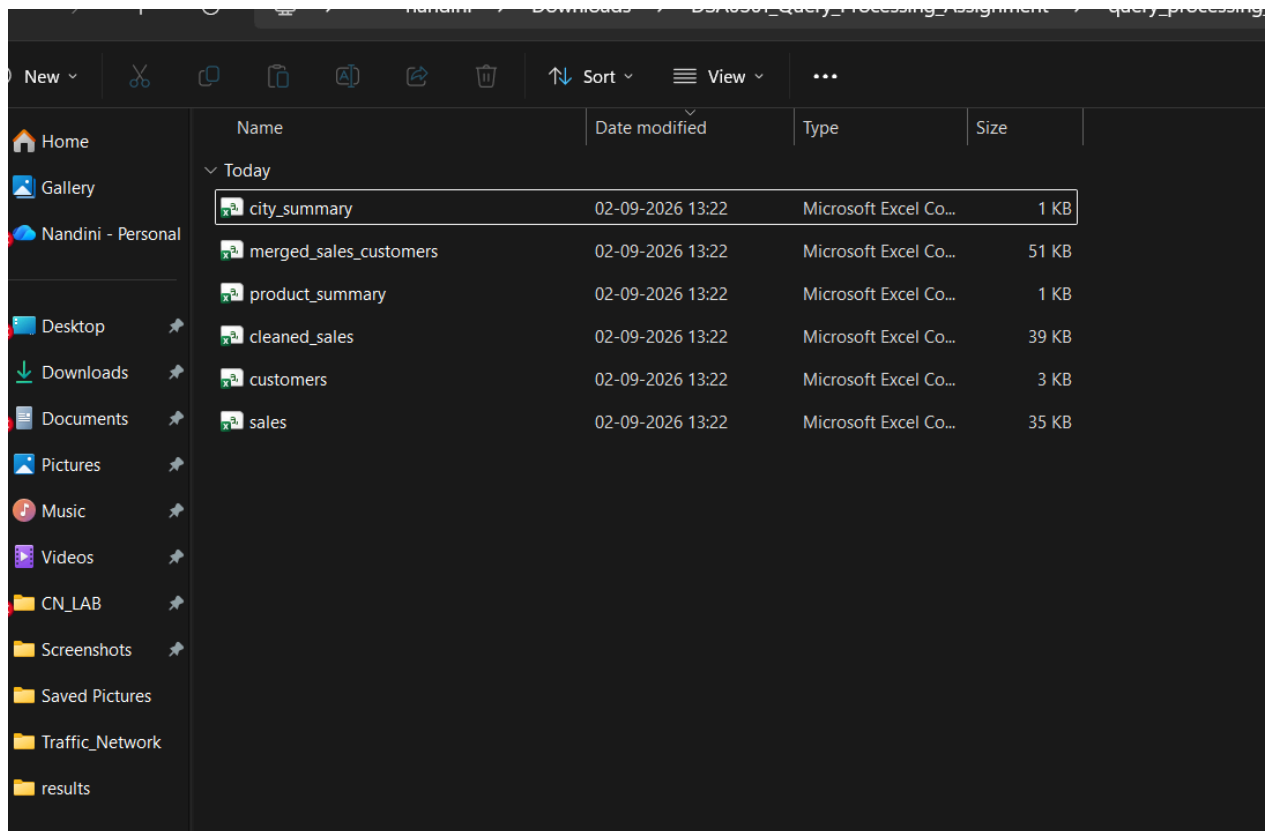


Figure 1: Dataset Files Used for Data Analysis

6. Data Importing and Exploration

Pandas is used to import the CSV datasets and inspect their structure. The `head()`, `shape`, `columns` and descriptive statistics functions are used to understand the dataset.

The dataset contains information about customers, products, transactions, prices, quantities, discounts and ratings.

Caption:

Figure 2: Dataset Importing and Initial Exploration

Paste under: Data Importing and Exploration.

7. Data Cleaning

Data cleaning is performed before analysis to improve the reliability of the results. Missing values in the **Rating** and **Discount_pct** columns are replaced using their median values.

The date column is also converted into a proper datetime format. Additional time-related features such as month, month name and day are extracted.

```
=== DATA IMPORTING AND EXPLORATION ===
Shape: (500, 10)
  Order_ID  Customer_ID    Date  ... Discount_pct  Rating  Sales_Amount
0         1         1084 2024-01-01  ...         3.27     5.0     155524.46
1         2         1020 2024-01-02  ...        10.97     2.5     57607.79
2         3         1081 2024-01-03  ...         7.57     3.6    124085.22
3         4         1001 2024-01-04  ...         9.23     4.6     31378.36
4         5         1080 2024-01-05  ...         5.90     3.3     21680.55

[5 rows x 10 columns]

Missing BEFORE:
  Order_ID      0
Customer_ID      0
Date            0
Product         0
Category        0
Quantity        0
Unit_Price      0
Discount_pct     2
Rating          3
Sales_Amount     0
dtype: int64

Missing AFTER:
  Order_ID      0
Customer_ID      0
Date            0
Product         0
Category        0
Quantity        0
Unit_Price      0
Discount_pct     0
Rating          0
Sales_Amount     0
Month           0
Month_Name      0
Customer_Name   0
City           0
dtype: int64
Bootstrap 95% CI: 130912.11 to 151312.54
```

Figure 3: Missing Value Handling Before and After Data Cleaning

8. Data Processing, Joining and Grouping

The sales and customer datasets are joined using the common **Customer_ID** column.

Sales Dataset

+

Customer Dataset



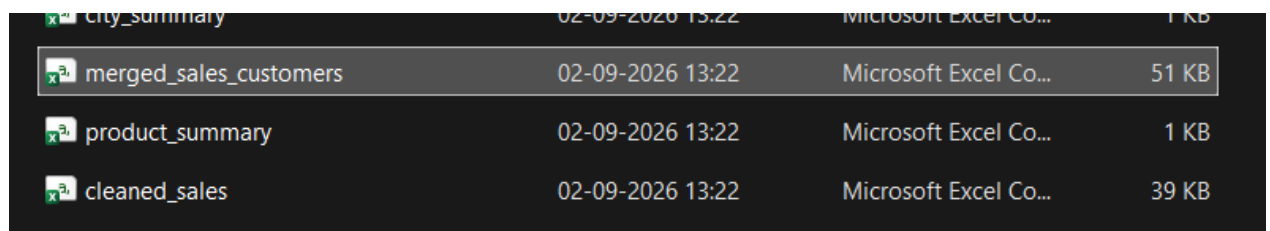
Merge using Customer_ID



Combined Dataset

After joining, the data is grouped based on city, product and category to calculate total sales, average sales and number of transactions.

Filtering is also performed to identify transactions whose sales amount is greater than the median sales value.



city_summary	02-09-2026 13:22	Microsoft Excel Co...	1 KB
merged_sales_customers	02-09-2026 13:22	Microsoft Excel Co...	51 KB
product_summary	02-09-2026 13:22	Microsoft Excel Co...	1 KB
cleaned_sales	02-09-2026 13:22	Microsoft Excel Co...	39 KB

Figure 4: Dataset Joining and Grouping Results

9. Statistical Analysis

Descriptive statistical analysis is performed using Pandas. Measures such as mean, standard deviation, minimum, maximum and quartiles are calculated for numerical variables.

The analysis helps understand the distribution and variation of quantities, prices, discounts, ratings and sales amounts

10. Correlation Analysis

Correlation analysis is used to identify relationships between numerical variables such as quantity, unit price, discount, rating and sales amount.

A correlation value close to +1 indicates a strong positive relationship, while a value close to -1 indicates a strong negative relationship. A value near zero indicates a weak linear relationship.

A	B	C	D	E	F	G
	Quantity	Unit_Price	Discount_	Rating	Sales_Amount	
Quantity	1	0.008003	-0.00485	-0.06099	0.618257	
Unit_Price	0.008003	1	-0.02185	-0.02282	0.698505	
Discount_	-0.00485	-0.02185	1	0.00125	-0.10061	
Rating	-0.06099	-0.02282	0.00125	1	-0.07563	
Sales_Amc	0.618257	0.698505	-0.10061	-0.07563	1	

Figure 6: Correlation Analysis of Numerical Variables

11. Outlier Detection

The Interquartile Range (IQR) method is used to identify unusual sales values.

The first quartile (Q1), third quartile (Q3) and IQR are calculated. Values outside the lower and upper limits are considered potential outliers.

The detected outliers are stored in:

results/sales_outliers.csv

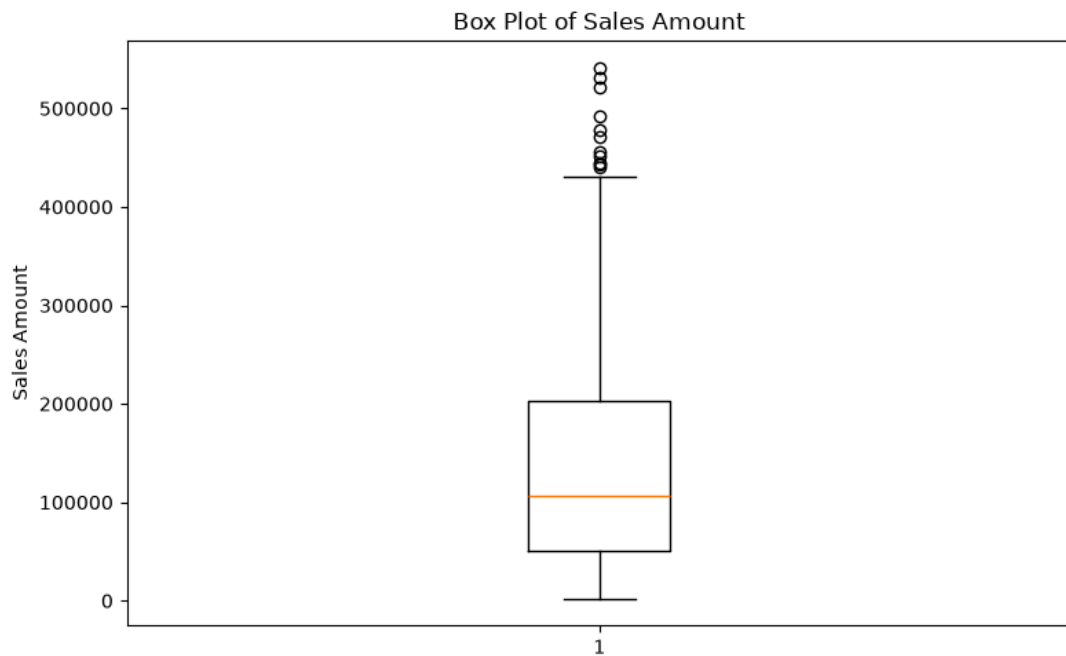


Figure 7: Box Plot for Identifying Sales Outliers

12. Data Visualization

Different visualization techniques are used to represent the analysed data.

12.1 Line Chart

A line chart is used to analyse the daily sales trend over time. It helps identify increases, decreases and fluctuations in sales.

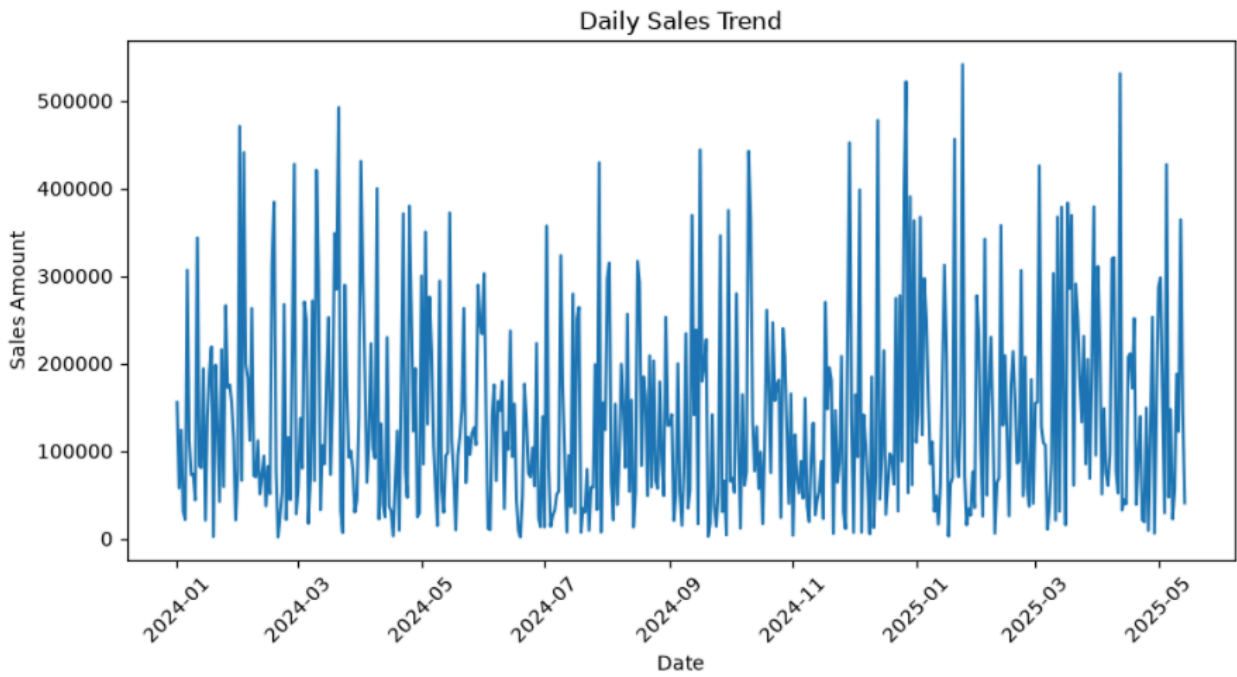


Figure 8: Daily Sales Trend Using Line Chart

12.2 Bar Chart

A bar chart is used to compare total sales between different products.

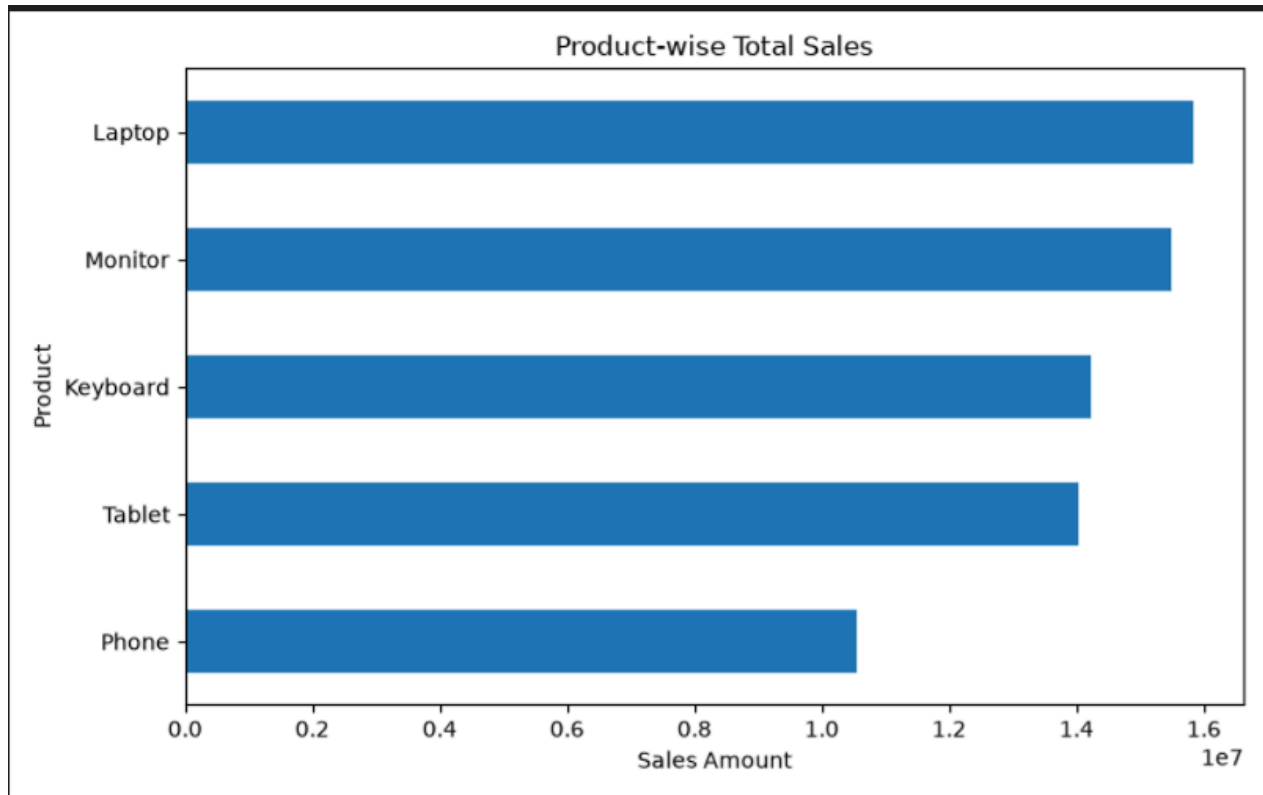


Figure 9: Product-wise Sales Comparison

12.3 Histogram

A histogram is used to study the distribution of sales amounts.

It helps identify the concentration and spread of the sales values.

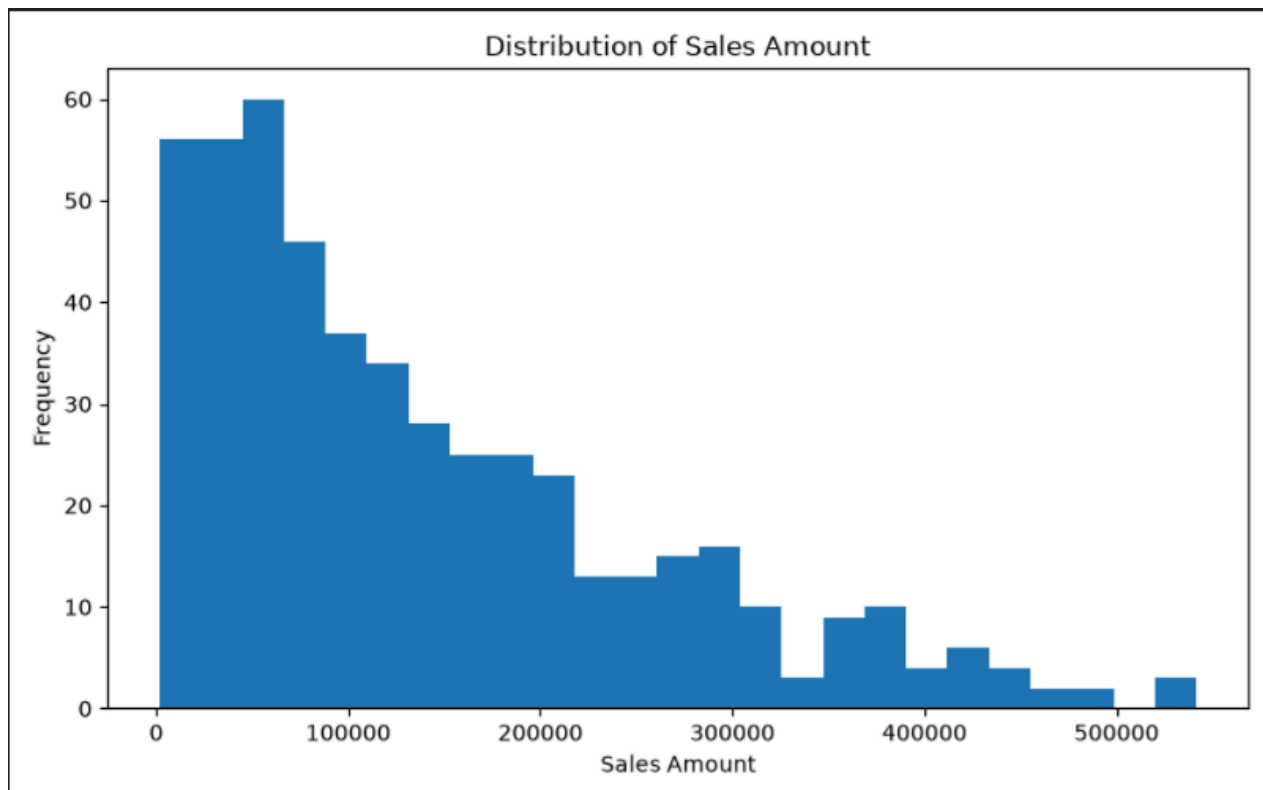


Figure 10: Distribution of Sales Amount Using Histogram

12.4 Scatter Plot

A scatter plot is used to study the relationship between unit price and sales amount.

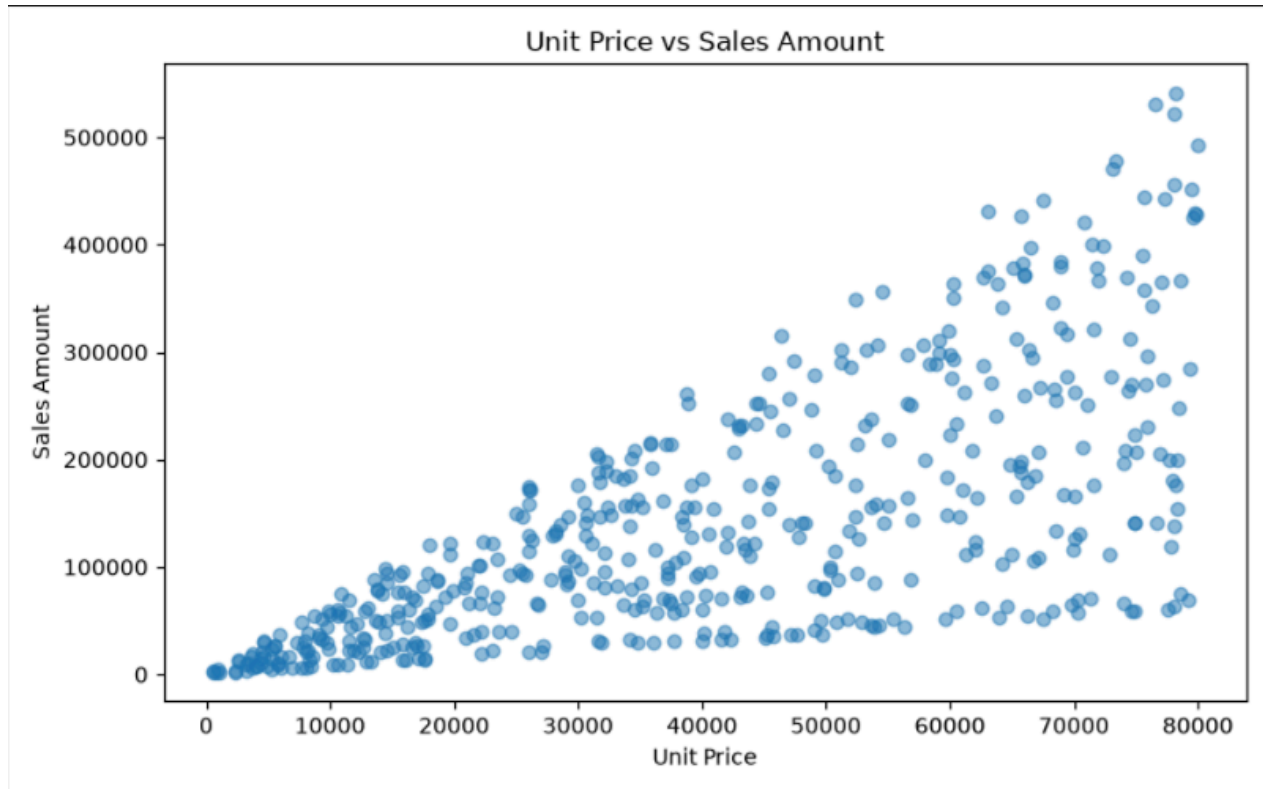


Figure 11: Relationship Between Unit Price and Sales Amount

13. Advanced Visualization Techniques

The assignment specifically requires advanced techniques including scatter matrices, lag plots, autocorrelation plots and bootstrap plots.

13.1 Scatter Matrix

A scatter matrix is used to visualize relationships among multiple numerical variables simultaneously.

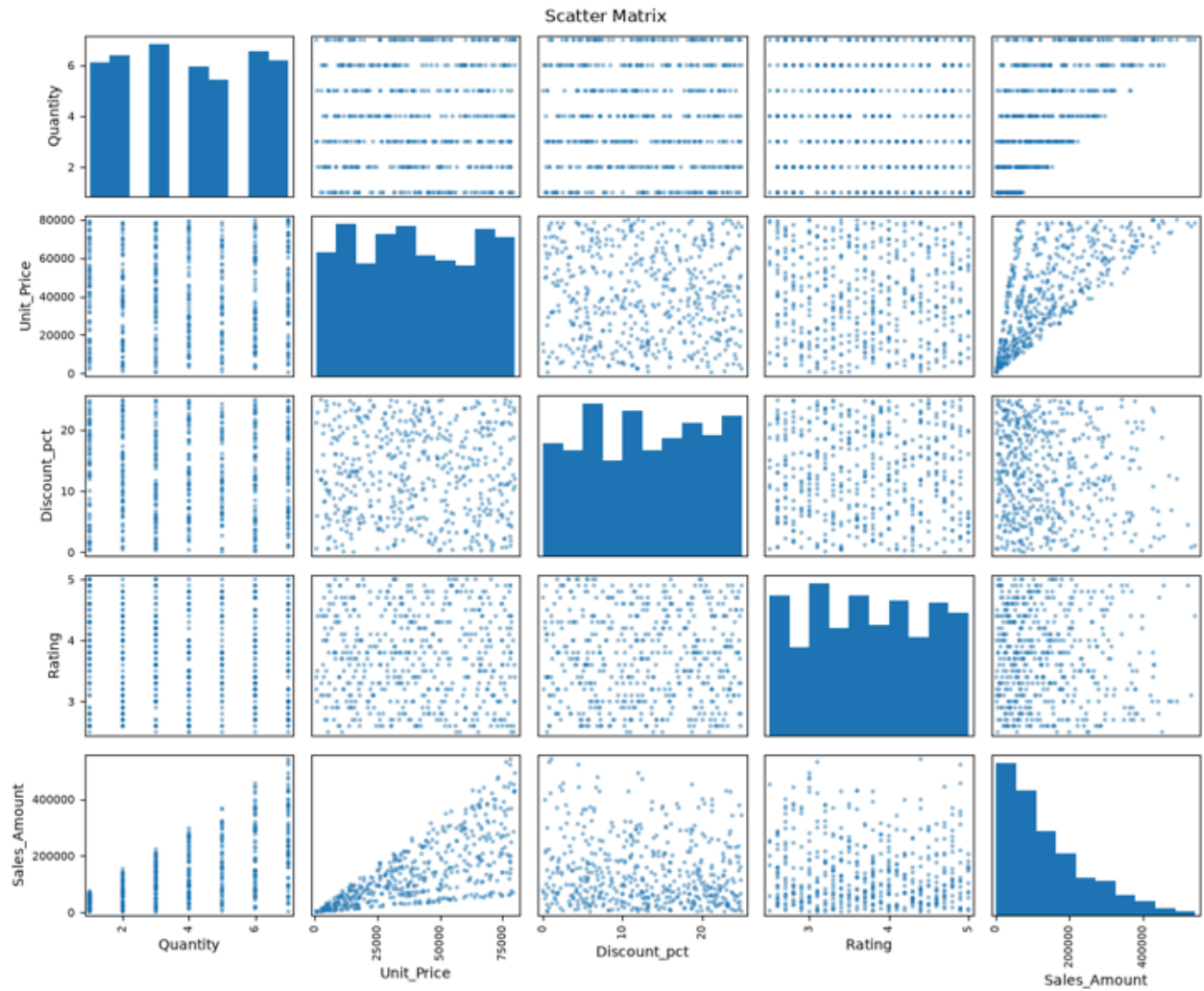


Figure 13: Scatter Matrix of Numerical Variables

13.2 Lag Plot

A lag plot compares a data value with its previous value. It can be used to investigate sequential dependence in sales data.

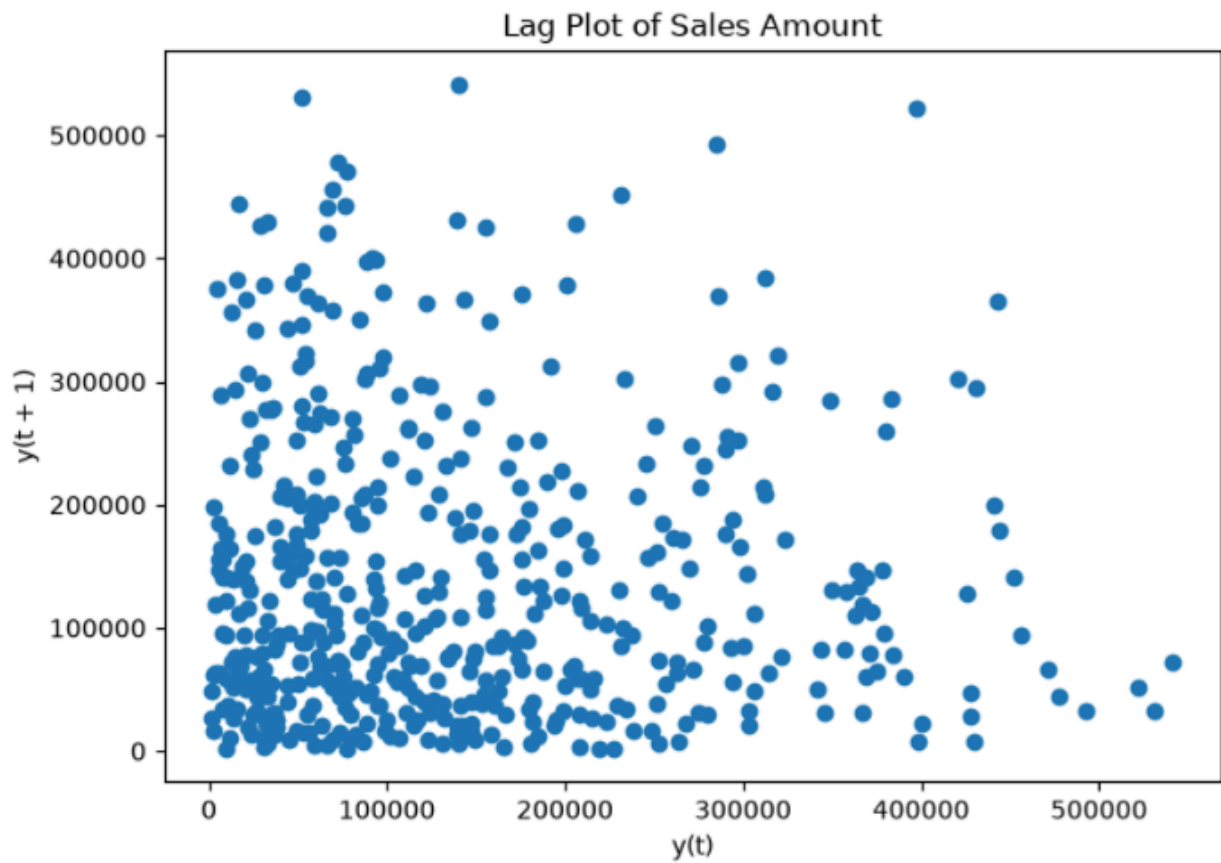


Figure 14: Lag Plot of Sales Amount

13.3 Autocorrelation Plot

Autocorrelation measures the relationship between a time series and its previous observations at different lag values.

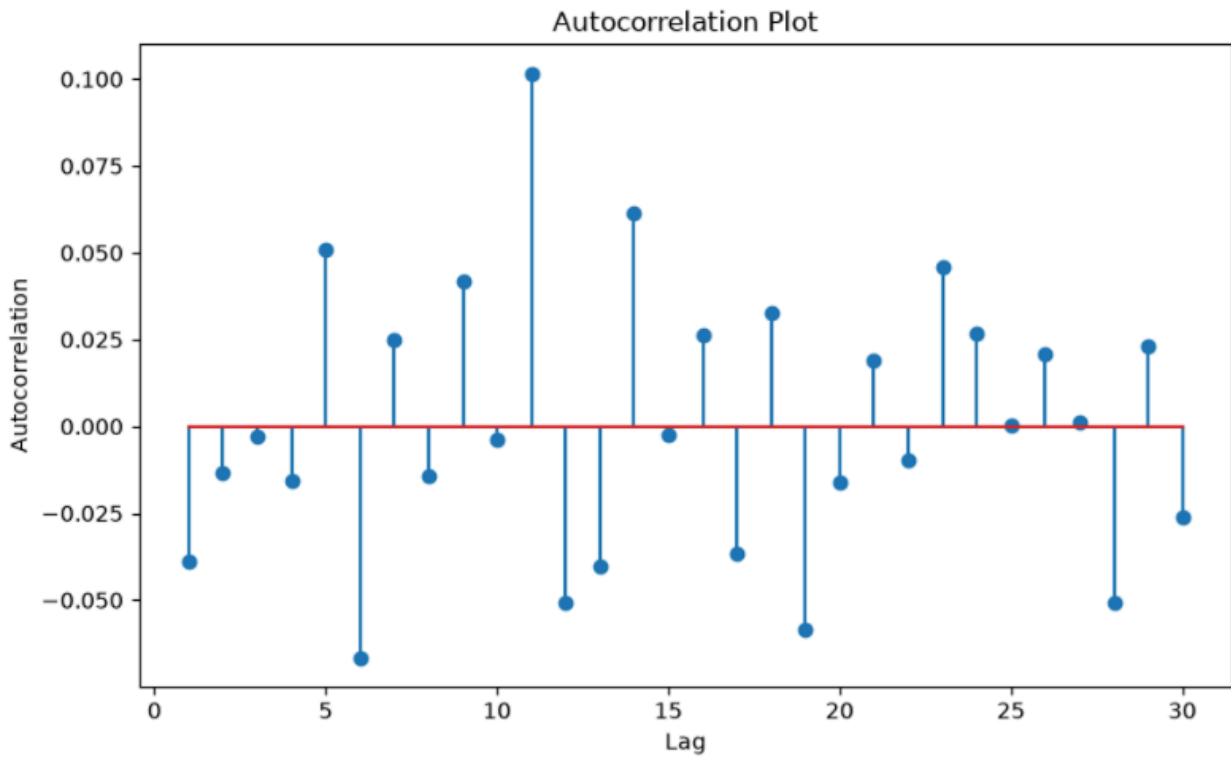


Figure 15: Autocorrelation of Sales Data

13.4 Bootstrap Plot

Bootstrap sampling is used to estimate the uncertainty of the mean sales value. Multiple samples are generated with replacement and their means are calculated.

A 95% confidence interval is obtained from the bootstrap distribution.

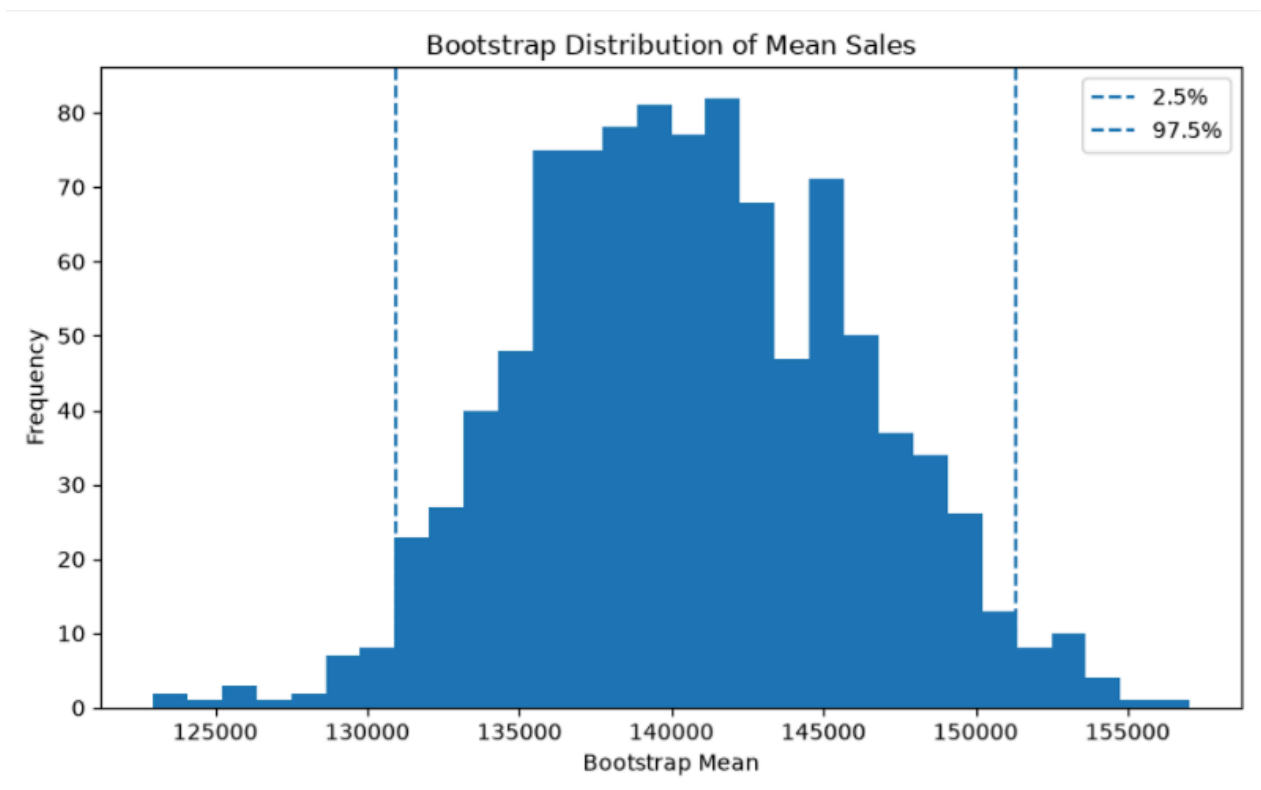


Figure 16: Bootstrap Distribution of Mean Sales

15. Visualization Technique Evaluation

Different visualization methods are suitable for different analytical requirements.

Visualization	Main Purpose	Suitability
Line Chart	Time trends	Very High
Bar Chart	Category comparison	Very High
Histogram	Distribution	High
Scatter Plot	Variable relationship	High
Box Plot	Outlier detection	Very High
Scatter Matrix	Multiple relationships	Medium
Lag Plot	Sequential	Medium

	relationship	
Autocorrelation	Time dependence	Medium
Bootstrap Plot	Uncertainty analysis	Medium

The evaluation demonstrates that no single visualization is suitable for every type of analysis. Line and bar charts are highly interpretable for general reporting, while scatter plots and box plots are useful for relationship and outlier analysis. Advanced plots provide deeper analytical information but may require more interpretation.

This matches the assignment's requirement to evaluate visualization techniques based on suitability, interpretability, pattern identification and practical applicability.

 Screenshot 18

	A	B	C	D	E
	Technique	Purpose			
	Line Chart	Time trends			
	Bar Chart	Category comparison			
	Histogram	Distribution			
	Scatter Plot	Variable relationship			
	Box Plot	Outliers			
	Scatter Matrix	Multiple relationships			
	Lag Plot	Sequential dependence			
	Autocorrelation	Time-series dependence			
0	Bootstrap	Uncertainty			
1					
2					
3					
4					

Figure 18: Evaluation of Visualization Techniques

16. Results and Findings

The analysis demonstrates that Pandas provides efficient methods for data importing, cleaning,

grouping, joining and statistical analysis. Matplotlib provides flexible visualization techniques for presenting analytical results.

The line chart helps identify time-based sales trends, while the bar chart provides an easy comparison between products. Histograms describe distributions, scatter plots reveal relationships and box plots help detect outliers.

Advanced techniques such as scatter matrices, lag plots, autocorrelation and bootstrap analysis provide additional information about relationships, sequential patterns and uncertainty.

17. Conclusion

This assignment provided practical experience in performing complete data exploration and visualization using Pandas and Matplotlib. The project demonstrated data importing, cleaning, joining, grouping, filtering, statistical analysis, correlation analysis and outlier detection.

Different visualization techniques were implemented to understand trends, distributions and relationships within the data. Advanced techniques such as scatter matrices, lag plots, autocorrelation plots and bootstrap analysis further improved the analytical understanding of the dataset.

Overall, the experiment shows that selecting an appropriate visualization based on the nature of the data is important for communicating insights clearly and supporting data-driven decision-making.

18. Student Reflection

What I Learned

I learned how to import, explore, clean and analyse datasets using Pandas. I also understood how different visualization techniques can reveal different patterns in data.

Techniques Learned

The most useful techniques were grouping, merging, filtering, correlation analysis and

visualization. I found bar charts and line charts particularly easy to interpret, while scatter plots and box plots were useful for understanding relationships and outliers.

Challenges Faced

One of the challenges was handling missing values and combining datasets using a common column. Understanding advanced visualization techniques such as autocorrelation and bootstrap analysis also required additional practice.

Data Insights

Correlation analysis helped identify relationships between numerical variables, while box plots helped identify unusual sales values. The visualizations made trends and distributions easier to understand.

Real-World Application

These techniques can be applied to business sales analysis, customer behaviour analysis, financial analysis, healthcare analytics and many other real-world data analytics problems.

Further Learning

In the future, I would like to work with larger real-world datasets and explore interactive visualization tools and machine learning techniques for predictive analytics.

S.No .	Assessment Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2–1)	Marks
1	Data Importing and Exploration	Correctly imports datasets and performs comprehensive data exploration using Pandas	Imports and explores datasets with minor errors	Performs basic data exploration	Limited or incorrect data exploration	5

2	Data Processing, Joining and Grouping	Effectively joins multiple datasets, creates groupings, filters and focuses data appropriately	Performs joining, grouping and filtering with minor issues	Performs basic joining and grouping operations	Incomplete or incorrect data processing	5
3	Data Analysis – Correlations and Outliers	Correctly identifies and interprets correlations, distributions and outliers with meaningful insights	Identifies correlations and outliers with reasonable interpretation	Identifies basic patterns with limited interpretation	Unable to correctly identify or interpret patterns	5
4	Visualization Using	Creates appropriate, well-labelled	Creates suitable charts with	Creates basic charts with limited	Visualizations are inappropriate,	5

	Pandas and Matplotlib	and meaningful charts using Pandas and Matplotlib with suitable	minor presentation issues	customization	incomplete or poorly labelled	
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		customization				
5	Advanced Visualization Techniques	Correctly implements and interprets scatter matrices, lag plots, autocorrelation plots and bootstrap plots	Implements most advanced plots with minor errors	Implements some advanced plots with basic interpretation	Limited or incorrect implementation and interpretation	5
6	Student Reflection and Learning	Clearly reflects on learning, challenges, insights gained and real-world application of the concepts	Provides meaningful reflection on learning and application	Provides basic reflection on the learning experience	Reflection is very brief, unclear or missing	5
Total						30

Reflection Area	Student Should Reflect On
What I Learned	What did I learn about data importing, exploration, analysis and visualization?
Techniques Learned	Which Pandas and Matplotlib techniques did I find most useful and why?
Challenges Faced	What errors or difficulties did I encounter while working with the datasets?

Data Insights	How did correlations, outlier detection and visualizations help me understand the data?
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Real-World Application	How can these techniques be applied to a real-world data analytics problem?
Further Learning	What would I improve, modify or explore further if given more time?